Personal details

- **Full name**: Robel K. Gebre, Ing, PhD
- **Address**: 4320 22nd Ave NW, Rochester Minnesota, 55901
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Employment details:

- **Name**: Mayo Clinic
- **Address**: 200 1st St SW, Rochester Minnesota, 55905
- **Date of employment**: 09/13/2021
- **Job Description:**
  - **Responsibilities**: Postdoctoral Research Fellow/Research Associate
  - **Completed projects:** Magnetic resonance imaging (MRI) scanner harmonization, novel tau positron emission tomography (PET) quantification, cognitive trajectory modeling from blood plasma biomarkers, Alzheimer's disease (AD) genetics using single nucleotide polymorphisms (SNPs) and machine learning, brain age prediction from MRI, and multiple system atrophy (MSA) signatures from structural and diffusion MRI.
  - **Active projects**: Early-onset prediction of MSA, clinical trial enrichment for AD anti-amyloid therapies, AD disease progression modeling, and cardiovascular disease (CVD) and cerebral amyloid angiopathy (CAA) profiling for anti-amyloid treatment (AAT).
  - **Accomplishments:** Novel tau PET metric, novel MSA imaging signatures, therapeutic window for AAT and clinical trials.

Degrees

- Doctor of Philosophy, Medicine (PhD)
  - Date awarded: 01 July 2021
  - Institution: University of Oulu, Oulu, Finland
  - Major/Program: Medical Physics and Technology
- Master of Science (MSc. Ing.)
  - Date awarded: 2016
  - Institutions: Joint Master's Degree Program at Czech Technical University (Prague, Czech Republic) and University of Groningen (Groningen, The Netherlands)
  - Major/Program: Biomedical Engineering and Clinical Technology
- Bachelor of Science (BSc)
  - Date awarded: 2013
  - Institution: Jimma University Institute of Technology, Jimma, Ethiopia
  - Major/Program: Biomedical Engineering

Awards and honours

- The Don Summers Memorial MSA Travel Fellowship Award (2025)
- Alavi-Mandell Award (2025)
- Young Investigator Travel Scholarship, Human Amyloid Imaging (2023)
- Mayo Clinic Radiology Research Fellowship, Mayo Clinic (2021)
- Newspaper covers on PhD thesis, Teknikka&Talous, Finland (2021)
- Orthopedic Research Society, Travel scholarship award (2019)
- Common European Master's Course in Biomedical Engineering Scholarship, Erasmus Mundus (2013)
- Summa cum laude, Jimma University (2013)
- Named best young scientist in Ethiopia for BSc project on "Wireless Infant Incubators"

Scientific and societal impact

- Invited talks:
  - University of Mississippi Medical Centre, The MIND Centre (2025)
  - Karolinska Institute Department of Neurobiology, Sweden (2024)
  - Mayo-Karolinska Institute Seminars (2022)
- PhD work was featured in two newspaper articles (2021):
  - Lonkkamurtumat vähentyvät mutta lantion murtumiset yleistyvät – taustalla luukato | Tekniikka&Talous
  - Dissertation: Machine learning methods succeeded in identifying fracture cases from controls and classifying hip osteoarthritis from CT images. | Kauppalehti

Publications (Selected)

1. **Gebre, R.K.,** Raghavan, S., De Tora, M.E.J. *et al* (2026). Precise disease heterogeneity and progression quantification in MSA and Parkinson's disease using machine learning. *Sci Rep* **16**, 10579. https://doi.org/10.1038/s41598-026-45949-5

2. Raghavan S, **Gebre R.K.**, Vemuri P. Chapter 8 - Artificial intelligence: Relevance and applications in dementia. In: Sharma CP, Thomas V, Nakhmani A, Smith RJ, eds. Artificial Intelligence and Brain-Computer Interfaces in Healthcare. Academic Press; 2026:143-153.
3. **Gebre R.K.**, et al. *Heterogeneity of MRI Biomarkers in Multiple System Atrophy and Parkinson's Disease Assessed Using Machine Learning*. Sep 2025. In Review. Available at SSRN: http://dx.doi.org/10.2139/ssrn.5403293.
4. **Gebre R.K.**, et al. *Advancing Tau PET quantification in Alzheimer disease with machine learning: introducing THETA, a novel tau summary measure.* Journal of Nuclear Medicine. 2024; DOI: https://doi.org/10.2967/jnumed.123.267273
5. **Gebre R.K.**, et al. *Can integration of Alzheimer's plasma biomarkers with MRI, cardiovascular, genetics, and lifestyle measures improve cognition prediction?* Brain Communications. 2024; DOI: https://doi.org/10.1093/braincomms/fcae300
6. **Gebre R.K.**, et al. *Cross–scanner harmonization methods for structural MRI may need further work: A comparison study.* NeuroImage. 2023; DOI: 10.1016/j.neuroimage.2023.119912
7. Gunter NB, **Gebre R.K.**, et al. *Machine learning models of polygenic risk for enhanced prediction of Alzheimer disease endophenotypes.* Neurology Genetics. 2024; DOI: **https://doi.org/10.1212/NXG.0000000000200012**
8. Ramanan VK, **Gebre R.K.**, et al. *Genetic risk scores enhance the diagnostic value of plasma biomarkers of brain amyloidosis.* Brain. 2023; DOI: https://doi.org/10.1093/brain/awad196
9. Raghavan S, Przybelski SA, Lesnick TG, Fought AJ, Reid RI, **Gebre R.K.**, et al. *Vascular risk, gait, behavioral, and plasma indicators of VCID*. Alzheimer's & Dementia. 2024; DOI: 10.1002/alz.13540
10. Tian J, Raghavan S, Reid RI, Przybelski SA, Lesnick TG, **Gebre R.K.**, et al. *White matter degeneration pathways associated with tau deposition in Alzheimer disease*. Neurology. 2023; DOI: 10.1212/WNL.0000000000207250
11. **Gebre R.K.**, et al. *Detecting hip osteoarthritis on clinical CT: a deep learning application based on 2-D summation images derived from CT.* Osteoporosis International. 2022; DOI: 10.1007/s00198-021-06130-y
12. **Gebre R.K.**, et al. *Structural risk factors for low-energy acetabular fractures*. Bone. 2019; DOI: 10.1016/j.bone.2019.07.004
13. **Gebre R.K.**, et al. *Discrimination of low-energy acetabular fractures from controls using computed tomography-based bone characteristics*. Annals of Biomedical Engineering. 2021; DOI: 10.1007/s10439-020-02563-4
14. Bhuyan M, **Gebre R.K.**, Finnilä M, et al. (2021). Preparation of filter by alkali activation of blast furnace slag and its application for dye removal. Journal of Environmental Chemical Engineering. 10. 107051. 10.1016/j.jece.2021.107051.
15. Raghavan, S., Przybelski, S. A., **Gebre, R. K.**, Low, A., Reid, R. I., Graff-Radford, J., & Vemuri, P. (2025). Diffusion Imaging Measures to Disentangle SVD-related Hypertensive Arteriopathy versus Cerebral Amyloid Angiopathy. *Cerebral Circulation - Cognition and Behavior, 9*, 100430. https://doi.org/10.1016/j.cccb.2025.100430
16. **Gebre, R. K.**, Przybelski, S. A., Raghavan, S., Gunter, J. L., Lowe, V. J., Graff-Radford, J., Knopman, D. S., Petersen, R., Jack, C. R., & Vemuri, P. (2025). Precise Estimation Of Heterogenous Global Tau Burden In The Brain Sheds Light On Cognitive Resilience Mechanisms. *Alzheimer's & Dementia, 21*(S2), e103992. https://doi.org/10.1002/alz70856_103992
17. Reid, R. I., **Gebre, R. K.**, Kamykowski, M. G., Senjem, M. L., Arani, A., Cogswell, P. M., Zeydan, B., Kantarci, O. H., Kantarci, K., Vemuri, P., & Jack, C. R. (2025). Automatic Classification of Head MR Series. *Alzheimer's & Dementia, 21*(S8), e109882. https://doi.org/10.1002/alz70862_109882
18. Reyes, C. D., **Gebre, R. K.**, Gunter, J. L., Hu, M., Schwarz, C. G., Meyer, N. K., Reid, R. I., Kantarci, K., Jack, C. R., & Vemuri, P. (2025). Harmonization Remains Unresolved: The Case of White Matter Hyperintensity Using ComBat And Deming Regression. *Alzheimer's & Dementia, 21*(S7), e108630. https://doi.org/10.1002/alz70861_108630
19. Satpathi, S., **Gebre, R. K.**, Gunter, J. L., Przybelski, S. A., Senjem, M. L., Kantarci, K., Jack, C. R., & Vemuri, P. (2025). Developing scanner change invariant brain age models for aging and dementia studies. *Alzheimer's & Dementia, 21*(S2), e097891. https://doi.org/10.1002/alz70856_097891
20. Carvalho, D., Gunter, N., **Gebre, R.K.**, Stuart, M., St Louis, E., Silber, M., Przybelski, S., Boeve, B., Graff-Redford, J., Jack, C., Peterson, R., & Vemuri, P. (2024). Mapping neuroimaging using artificial intelligence to detect hypersomnia and its neurobiological correlates. *Sleep Medicine, 115*, S76–S77. https://doi.org/10.1016/j.sleep.2023.11.242
21. **Gebre, R. K.**, Jack, C. R., & Vemuri, P. (2024). Identifying transition points in AD biomarkers using machine learning: A supplement to the reliable worsening method. *Alzheimer's & Dementia, 20*(S9), e093902. https://doi.org/10.1002/alz.093902
22. **Gebre, R. K.**, Moscoso, A., Raghavan, S., Wiste, H. J., Heeman, F., Costoya-Sánchez, A., Schwarz, C. G., Spychalla, A. J., Lowe, V. J., Graff-Radford, J., Knopman, D. S., Petersen, R. C., Schöll, M., Murray, M. E., Jack, C. R., & Vemuri, P. (2024). Validation Of the Tau Heterogeneity Evaluation in Alzheimer's Disease (THETA) Score Using Longitudinal and Histopathology Data. *Alzheimer's & Dementia, 20*(S1), e091770. https://doi.org/10.1002/alz.091770
23. Yu, X., Przybelski, S. A., Reid, R. I., **Gebre, R. K.**, Lesnick, T. G., Vassilaki, M., Graff-Radford, J., Knopman, D. S., Petersen, R. C., Jack, C. R., & Vemuri, P. (2024). Chronic Medical Conditions and Dementia Risk: Brain Age Models for Quantifying Impact and Understanding Mechanisms. *Alzheimer's & Dementia, 20*(S7), e089382. https://doi.org/10.1002/alz.089382

24. Raghavan, S., Przybelski, S. A., Lesnick, T. G., Fought, A. J., Reid, R. I., **Gebre, R. K.,** Windham, B. G., Algeciras-Schimnich, A., Machulda, M. M., Vassilaki, M., Knopman, D. S., Jack, C. R., Petersen, R. C., Graff-Radford, J., & Vemuri, P. (2023). Vascular risk, gait, behavioral, and plasma indicators of VCID. *Alzheimer's & Dementia*, alz.13540. https://doi.org/10.1002/alz.13540
25. Raghavan, S., Przybelski, S. A., Lesnick, T. G., Reid, R. I., Fought, A. J., **Gebre, R. K.,** Algeciras-Schimnich, A., Machulda, M. M., Vassilaki, M., Knopman, D. S., Jack, C. R., Petersen, R., Graff-Radford, J., & Vemuri, P. (2023). Non-imaging measures to predict variability in white matter changes relevant to VCID. *Alzheimer's & Dementia, 19*(S17), e077200. https://doi.org/10.1002/alz.077200
26. Yu, X., **Gebre, R. K.,** Przybelski, S. A., Reid, R. I., Raghavan, S., Lesnick, T. G., Lowe, V. J., Machulda, M. M., Knopman, D. S., Petersen, R. C., Kantarci, K., Graff-Radford, J., Jack, C. R., & Vemuri, P. (2023). NODDI May Be More Sensitive to Neurodegenerative Changes Than Structural MRI. *Alzheimer's & Dementia, 19*(S17), e076509. https://doi.org/10.1002/alz.076509

Research grants awarded

- Active Grants
  - Investigating Resistance and Resilience Mechanisms in Alzheimer's Disease. Funded by National Institute on Aging. (FP00090851.02-PR08) – Collaborator
- Completed Grants
  - Radiology Research Fellowship. Funded by Mayo Clinic Radiology – Recipient

Novel methods and software developed

- **Two-Way Epistatic Interactions**: "A Novel Approach to Encode Two-Way Epistatic Interactions Between Single Nucleotide Polymorphisms" arXiV, Cornell University (Jun 2023): DOI: https://doi.org/10.48550/arXiv.2306.09175
- **Transition Point Estimator (TPE):** Alzheimer's Imaging Consortium (2024): "Identifying transition points in AD biomarkers using machine learning: A supplement to the reliable worsening method" GitHub: https://github.com/RobelGebre/TPE-for-SHAP
- **THETA:** A Novel Tau Heterogeneity Evaluation in Alzheimer's Disease. GitHub: https://github.com/RobelGebre/THETA

Peer reviewed journals

- Artificial Intelligence in Medicine
- Insights into Imaging
- Nature Communication
- Neurobiology of Aging
- NeuroImage
- Osteoporosis International
- Bone
- PlosOne

Professional memberships

- American Autonomic Society (AAS)
- AAIC ISTART